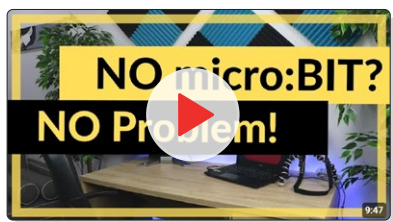
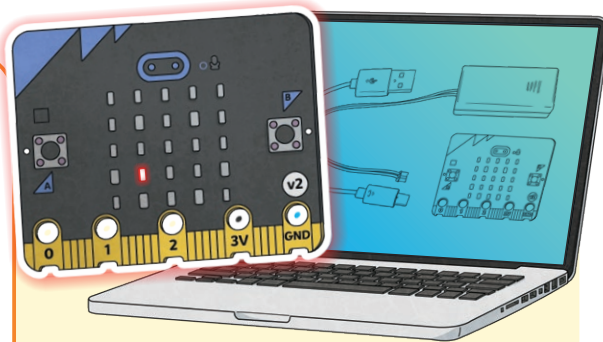


Introduction to MicroBits

Activity 1



- Introduce the learners to Microbits with the video.

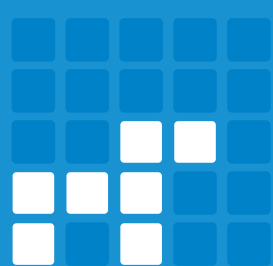


Activity 2

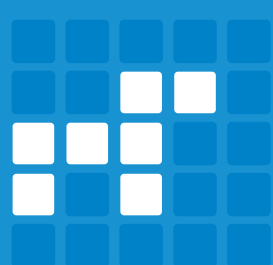
Write code to display the following images using different inputs. (Button A pressed, Button B pressed and Button A+B pressed.)

on button A pressed

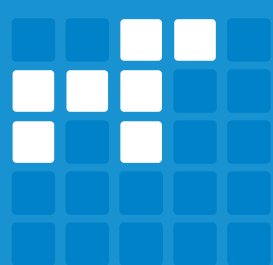
show leds



show leds

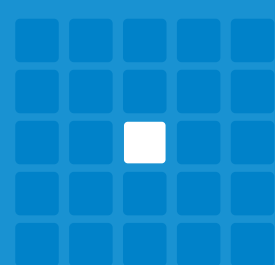


show leds



on button B pressed

show leds



show icon



show icon



show icon



show icon



on button A+B pressed

show icon



show icon



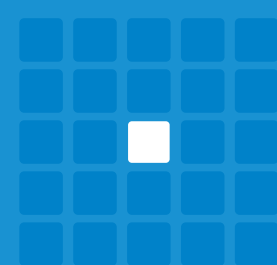
show icon



show icon



show leds



Disclaimer: We hope you find the information on our website and resources useful. This resource contains links to external video websites. These websites often have autoplay features meaning that other videos will play after the video you are watching finishes. You should disable this feature before using the video in any classroom or similar setting. Twinkl assumes no responsibility for the contents of linked websites. The inclusion of any link in this resource should not be taken as an endorsement of any kind by Twinkl of the linked website or any association with its operators. We have no control over the availability of the linked pages. If the link is not working, please let us know by contacting TwinklCares and we will try to fix it, although we can assume no responsibility if this is the case.

Activity 3

Input, sequence, and basic loop. Write code to display the following images using different inputs.

```

on button A pressed
  repeat 3 times
  do
    show leds
    [5x5 grid of white LEDs]
    pause (ms) 100
    show leds
    [3x3 grid of white LEDs]
    pause (ms) 100
    show leds
    [3x3 grid of white LEDs]
    pause (ms) 100
  
```

```

on button B pressed
  repeat 3 times
  do
    show leds
    [1x1 grid of white LED]
    pause (ms) 100
    show icon
    [5x5 grid of white LEDs]
    pause (ms) 1000
    show icon
    [3x3 grid of white LEDs]
    pause (ms) 1000
    show icon
    [3x3 grid of white LEDs]
    pause (ms) 1000
    show icon
    [5x5 grid of white LEDs]
  
```

```

on start
  show string "Hello!"
  
```

```

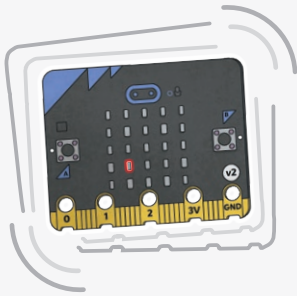
on button A+B pressed
  repeat 3 times
  do
    show icon
    [5x5 grid of white LEDs]
    show icon
    [3x3 grid of white LEDs]
    show icon
    [3x3 grid of white LEDs]
    show icon
    [3x3 grid of white LEDs]
    show leds
    [1x1 grid of white LED]
  
```

Activity 4

Write code to display the following images using different inputs.

on start

show string "YES"



on shake

if pick random 0 to 10 > 1 then

show icon

pause (ms) 100

show icon

pause (ms) 100

show icon

pause (ms) 100

show icon

clear screen

else

show string "o"

show icon

show icon

show icon

show icon

clear screen

Activity 5

Write code to display the following images using different inputs.

```

on button A pressed
  repeat 2 times
    do
      show icon [3x3 grid]
      play melody [melody] at tempo 120 bpm until done
      pause (ms) 500
      clear screen
      pause (ms) 100

on button B pressed
  repeat 2 times
    do
      show icon [3x3 grid]
      play melody [melody] at tempo 120 bpm until done
      pause (ms) 100
      clear screen
      pause (ms) 500

on button A+B pressed
  repeat 2 times
    do
      show icon [3x3 grid]
      play melody [melody] at tempo 120 bpm until done
      pause (ms) 500
      clear screen
      pause (ms) 500
  
```